Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · UE25CV141A Unit 3 Class 54 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE25CV141A Unit 3 Class 54 AV Summary

1 UE25CV141A Unit 3 Class 54 AV Summary

Study focus: UE25CV141A_Unit 3_Class 54_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SUV Vinkitish from Civil Engineering under the Department of Science and Humanities, PES University.
- This is class number 54.
- We are in Unit three, Distributed Forces.
- And chapter is Distributed Forces.
- Topic is beams internal effect.
- In the last class, that is class number 53, we had discussed about a problem that is phi bar 125.
- So first problem that I've chosen for today's class is phi bar 126.
- The statement of the problem goes like this.
- Determine the shear force V at a section B between A and C.
- And the bending movement at the support A.
- The point load is acting at the free end of the beam of magnitude 1.4 kilonewton and it is acting downwards as you can see.
- Then we have one more point load, which is acting at a distance of 1.5 meters from the free end, that is from the left end.
- of magnitude 1.8 kilonewtons.
- Then we have a roller support at point A, which is at a distance of 1.6 meters from the lower point 1.8 kilonewtons.
- And you have one more support at point C.
- Which is at a distance of 1.8 meters from support A.

- If you observe the support A, as I have told you, it is a roller support, and the support at C is a hinge or a pin support.
- So, first thing that we have to do here is to determine the support reactions as we have to determine the shear force at B.
- Whether we take to the left side of the section or to the right side of the section, B, we need to determine the support reaction at A.
- And C to find out or to take the section to the right side of the beam.
- So to first to find out the support reaction, as we have explained in beam's internal effect, we have to use the three equations of equilibrium.
- The free body diagram looks like this.
- What do we have in the free body diagram is the point load of one point four kilometers is acting downwards at the free end, that is the left side end.
- Then one more load that is of one point eight kilonewtons, which is acting at a distance of one point five meters from the left end.
- Then we have the support at A which is being replaced by its reaction R A, since it is a roller support, the reaction will be only one and it is acting upwards and it is denoted as RA.
- Then we have a support at C which is a pin support or a hint support, so we will have two reactions, one vertical, which is denoted as V C and one horizontal, which is denoted as H C.
- For which we will use the equations of equilibrium.
- So we will take first equation of equilibrium as summation of the moment to be equal to 0, and I have taken the moment at point C because we have two reactions acting at this point.
- We get this point load.
- Which is 1.4 kilometer into the perpendicular distance is totally equal to 1.5 plus 1.6 plus 1.8.
- That is what is indicated here, or you can take it from C, which is 1.8 plus 1.6 plus 1.5, and that's what is indicated here.
- Point load that is one point four kilometer will produce an anti clockwise movement, so we will take it as positive.
- Which is also a positive because it is acting in the anti-clockwise direction.
- Then we have one more point load that is acting because of the support at A, which is upwards, which will create a clockwise moment at C.

Example or application from the lecture

- The diagram, if you read the diagram, what is given in the diagram is there is a point load that is acting.
- So first let us draw the free body diagram.
- So first let's find out the values of RA, V C and H C.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

And that is what we have the forces acting on this beam.

Key details

- Next, to find out the other support that is V C, we will take all the summation of all the forces, algebraic sum of all the forces acting along the y direction to be equal to 0.
- But one R A which is equal to 7.2 kilon, which is acting upwards, which will be positive.
- Which is equal to minus 4.01 kilonewt, which indicates that since there is a negative sign here.
- That indicates that the reaction at C is not acting upwards, it is acting downwards.
- And that is possible because we have a support at C as an heat support, so we can have either upwards reaction or downwards reaction.
- I hope that is clear.
- Next we will take the other equation of equilibrium that is algebraic sum of all the forces acting along the x direction to be equal to 0.
- So we will get HA to be equal to 0 because there is no horizontal force acting on the beam, so H will be equal to 0.
- What is asked in the problem is to determine the shear force at a section B, that is what is indicated here, which is just to the right side of A.
- However, here in this problem I have shown you by both, whether you take to the left or to take to the right, using the sign convention that was mentioned in the last class, you will get the same value for the shear force.
- That is what I've indicated and that is why.
- You can see the free body diagram that is written here, where I have taken only the beam acting to the left side of the section beam.
- This is what is the beam body diagram.
- We have only VC and what we have got for V C is negative, which indicates that it is acting downwards, so that is what is indicated here.
- Next, they have also asked us to determine the movement about A.
- Similar to the shear force, I have also I have written the diagram of the beam.
- Which is to the left side of A and right side of A, that is this one.
- We have point load one point four and one point eight, both of them acting downwards.
- And at A we also have a reaction at A R A.
- Then if you take to the right side of the section, again we have a point load or the reaction at V C that is acting, which is downwards, which is four point zero one, and the reaction at A that is acting.
- Going forward to find out the shear force at V here, if I take the section to be acting to the.
- Left side or the beams acting to the left side, what do we get?
- We get 1.4 acting downwards, which is negative.
- 1.8 is also acting downwards to the left side of the section, so it is negative.
- Then we have.
- RA which is acting to the left side of the section upwards which is positive.

Example or application from the lecture

- When we consider to the right side of the section.
- Whatever value that we get, that should be the same value what we get it when we consider to the right side of the section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 10:00–14:59

Let's see that.

Key details

- Then coming to the bending moment, we take the bending movement at this point.
- Whatever I have discussed, the beam taken to the left side of the section B.
- Shear force at B will be equal to 1.8, which is acting downwards.
- Then we have 1.8 which is acting downwards, 1.4 is also acting downwards, both of them is indicated here.
- Plus RA which is acting upwards, so that is 7.21.
- Considering to the right side of the section, that is this one, we get the beam acting to the right side and the point rod is acting downwards.
- Next, to find out the bending moment, similarly, take the section to the left side from A.
- 1.6 is the distance, that is this distance, plus 1.5, so that is what we have written here.
- one point eight, which will also be hanging.
- into 1.6 so we get the value as 7.22.
- This is with respect to when we take the beam to the left side of the section.
- We get 4.01 into this distance of 1.8 which will also be Augin.
- Minus 7.219.
- That is what I have proved here for shear force as well as petting movement.
- I hope you have understood this problem.
- The problem statements go like this.
- Draw the shear and bending movement diagrams for the diving board, which supports the ATKG man poised to dive.
- Specify the bending moment with the maximum magnitude.
- In the diagram, the man is acting at the free end of the board.
- And also we should specify the maximum winning moment.
- And also the distance between A and B is one point two meters, whereas the distance between A to the free end of the beam or the board where the person is standing is four meters.
- So, first we have to find out the support reactions for which we will take the equations of equilibrium, considering summation of all the moments acting at A to be equal to 0.
- We will get when we take moment about A, we will get the weight of the person, which is 18 to 9.81, and the distance from A to the point where the mo where the person is.
- Is acting or the force of the person is acting is 4 meters, so that is 4 here.
- That would produce a clockwise movement.

Example or application from the lecture

- Next, let's go to the next problem, which is 5 bar 128.
- The second convention used is given here, clockwise is positive.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then we will take summation of all the forces acting upwards to be equal to 0.

Key details

- Of course it is indicated that the reaction is acting downwards, but what is taken here is that it is acting upwards, that is why it is positive, which is equal to zero.
- But once we calculate the value, we get R1 to be equal to negative, and that is why this is indicated as downwards.
- Remember that this R1 is positive because we are assuming it to be acting upwards.
- What we can do is.
- You write down horizontal lines under the point loads plus the support reactions, that's what is indicated here.
- Then draw a datum, draw a datum line, which indicates a 0, 0.
- You go down as the arrow of the support reaction is indicated, which is downwards of R1, then there is no load, so horizontal line.
- Then at B we have a reaction acting upwards, which is 26.
- You take it from the horizontal line above, so you get a value of 785.
- That is 2650 minus 1875.
- The point where the point load is acting to the edge of the beam, there is no load, so horizontal then go down by H and 5.
- For bending moment, determine the moment at B either taking the section to the left side or right side.
- And then draw the bending moment diagram.
- I hope all of you have understood.
- Now next we have asked you to find out the maximum bending moment in the beam, which will occur at the point where the support is acting at B that is equal to the R1 into the distance of.
- A to B.
- I hope all of you have understood this.
- We are going to be stopping this class here.
- My name is Dr.
- Sv Mektesh from civil engineering and the department of.
- Science of Minnesota, PS University.
- See you in the next class.
- That is class number 55.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE25CV141A_Unit 3_Class 54_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.

2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice